



Enhancing credit risk prediction with hybrid deep learning and sand cat swarm feature selection

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Received: 17 April 2023 / Revised: 20 September 2023 / Accepted: 21 December 2023 /

Published online: 4 January 2024

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Abstract

Credit risk prediction method acts as a vital financial tool for measuring the default probability of credit applicants. For financial institutions, proper credit risk management becomes mandatory to avoid significant losses incurred by borrowers' default. Thus, statistics are an increasingly vital technique that can analyse and measure credit risk. Generally, manual auditing and statistical methods measure credit risk. Current developments in financial artificial intelligence (AI) evolved from machine learning (ML)-driven credit risk methods that obtained great interest from academia and industry. The most significant step in the process of developing a credit risk assessment method is feature selection, which chooses a subset of appropriate features for enhancing the performance of an ML technique. With this motivation, this study concentrates on the design of sand cat swarm optimization-based feature selection with hybrid deep learning (SCSOFS-HDL) model for credit risk assessment. The presented SCSOFS-HDL technique presents a new SCSOFS technique for the optimal selection of feature subsets from the credit risk data. In addition, the deep LSTM Supervised Autoencoder Neural Network (DLSTM-SANN) model is presented for classification purposes. To enhance the performance of the DLSTM-SANN technique, the political optimizer (PO) methodology is utilized for the hyperparameter tuning process. The experimental validation of the SCSOFS-HDL technique is tested on credit risk datasets and the results highlighted the better performance of the SCSOFS-HDL algorithm with maximum accuracy of 96.49% and 96.12% on German Credit and Australian Credit datasets, respectively.

Keywords Credit risk analysis · Feature selection · Autoencoder · Machine learning · Political optimizer

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1 Introduction

Credit risk management is important for financial institutions or lending institutions whose core activity is lending [1]. Hence, corporation credit assessment or precise consumer is of utmost significance as financial entities probably incur significant losses if borrowers default [1]. Thus, finance companies must have a proper assessment regarding the credit risks of borrowers to control losses [2]. So, they strive to gather data from the borrower, and a lot of statistical approaches were developed to analyze and measure credit risk [3]. To predict the default possibility of an applicant as per the demographic and personal information of the applicants, the credit risk prediction method is used as the main tool [4]. The loan qualifications of the applicants are judged mainly by expert experience from previous decisions. On the other hand, with the rapid growth of artificial intelligence (AI) and big data, historical data from applicants were highly demanded to learn an advanced classification method for the task of credit risk prediction because of the assumption that the credit status or CIBIL score of a borrower was depicted by historical behaviour [5].

The task named credit risk prediction refers to a common binary classification issue that splits the population of borrowers into two they are a ‘bad’ class and a ‘good’ class [6]. A lot of studies have inspected dichotomous prediction methods to forecast the applicant’s probability of default [7]. Machine learning (ML) is now been added to the list of most effective techniques and comparatively achieves superior outcomes than generalized linear techniques [8] like support vector machines (SVM), decision trees (DT), gradient boosting (GB), neural networks (NN) and random forest (RF) which are used in real-time [3]. From the current literature, an analysis of a trend shows that the method competing for the maximum prediction accuracy are deep neural networks (DNN) or Deep Learning (DL) and gradient boosting machines (GBM) [9]. But, to use the classification data in the feature set and to weaken the complicated nonlinear relationship between the features, feature engineering becomes a vital task before building the credit risk prediction method. Feature engineering executes a sequence of engineering processes on the original dataset and refines them into features, which are utilized as inputs for methods [10]. It includes feature transformation, feature selection, and feature extraction [11]. Feature selection boosts the prediction performances of the method by removing redundant or irrelevant features to identify a feature subset that has a strong discriminative capability [12].

This study focuses mainly on the design of sand cat swarm optimization based feature selection with hybrid deep learning (SCSOFS-HDL) model for credit risk assessment. The presented SCSOFS-HDL technique presents a new SCSOFS technique for the optimal selection of feature subsets from the credit risk data. In addition, the deep LSTM Supervised Autoencoder Neural Network (DLSTM-SANN) model is presented for classification purposes. To improvize the performance of the DLSTM-SANN method, the political optimizer (PO) algorithm is utilized for hyperparameter tuning process. The experimental validation of the SCSOFS-HDL technique is tested on credit risk datasets.

2 Related works

The authors in [13], designed a composite index for the effect of financial volatility in uncertainty conditions in the financial marketplace and measured the Iranian financial system and the Tehran Stock Exchange from 2008 to 2020. The three key elements

in the design of this technique are banking sector, Volatility Index of currency, and stock exchange. Meanwhile, the earlier research used the shock of variables. This study considered a stock exchange, banking sector, and comprehensive index regarding the volatility of currency while utilizing the volatility of such variables. In [14], the authors developed an OALOFs-MLC technique (an oppositional ant lion optimizer-based feature selection with an ML-assisted classification) method for FCP in big data. The Hadoop MapReduce tool had been utilized for big data management in field of finance. On top of that, to filter an optimum set of features aids to get enhanced classification outcomes, the OALOFs-MLC approach creative new OALOFs technique was designed. Besides, to complete the grading process, the deep random vector functional links network (DRVFLN) method was utilized. With a baseline dataset, series of experiments of the OALOFs-MLC method was directed and the outcomes were proven as pre-dominant methods compared to other methods.

Elhoseny et al. [15] presented a political optimizer-based deep neural network (OD-PODNN) method, which is an outlier detection technique for financial crisis prediction. The presented techniques conclude the financial status of a company or business by numerous processes like classification, pre-processing, hyperparameter optimization, and outlier detection. The isolation forest (iForest) based outlier detection technique is used by the OD-PODNN. Also, the DNN hyperparameter was fine-tuned and the PODNN-based is derived to further enhance the overall classification accuracy. Three datasets were utilized, and the results are scrutinized under different performance measures, for assessing the OD-PODNN technique. Among other methods, the outcomes have proved the supremacy of the OD-PODNN approach. Park and Yang [16] presented government debt in emerging countries and private debt in developed nations developed as leading factors that hinder future economic growth. Concerning the COVID-19 impact, the author found that expansion of government debt and sharply reduced interest rates augmented the chance of a crisis in a few developing countries in the future.

In [17], an adaptive whale optimization algorithm with DL (AWOA-DL) approach is developed for making a novel financial distress approach. The AWOA-DL algorithm determines whether a firm is undergoing financial problems or not. Mainly, the DNN method obtains financial information as input and forecasts financial problems. Besides, to tune the DNN hyperparameters here applied the AWOA increases the prediction outcome. The authors in [18] compared the efficiency of DL method with other ML approaches like support vector machine (SVM) and random forest (RF) along with conventional statistical techniques like the linear regression method. In [19], the authors present a GMDH-type neural network technique that has outpaced the orthodox ML approaches, which achieves reliable outcomes of predictive modelling and testing hypotheses. Since a preferred network of near-optimal connectivity is studied from dataset, the abovementioned method modelling does not necessitate network structure knowledge. The outcomes compared with regard to predictive error display that GMDH-type predictive method has lesser errors compared to classical neural network and autoregressive methods.

In [20], the authors forecast commercial customer credit scores utilizing hybrid ML approaches that integrate supervised and unsupervised ML approaches. The authors execute distinct methods and relate to the efficiency of the hybrid approaches to that of separate supervised ML methods. In [21], a Wasserstein GAN (WGAN)-based data augmentation and hybrid FS approach has been presented for smaller instance credit risk assessment with higher dimensionality. Liu et al. [22] establish a 2-stage hybrid approach for enhancing the prediction outcome of credit risk. Initially, to make entire utilize of classified data hidden in credit data, the authors utilize XGBoost for linearizing and transforming the new features into a high-dimensional sparse feature matrix.

While there has been significant progress in the field of credit risk prediction using ML and DL models, there exists a notable research gap relating to the optimization of FS and hyperparameter tuning processes. Several credit risk prediction models available in the literature tend to focus on the selection of specific ML algorithms and the choice of input features without giving adequate attention to the crucial steps of FS and hyperparameter tuning. This research gap is particularly evident in the context of complex, high-dimensional datasets often encountered in credit risk assessment, where the relevance of features and the optimal configuration of model hyperparameters can greatly impact the predictive performance and interpretability of the models. Addressing this gap is vital as it can lead to more accurate and robust credit risk prediction models that are essential for financial institutions to make informed lending decisions while minimizing default risks. Therefore, this study develops a credit risk assessment model with the inclusion of FS and hyperparameter tuning process.

3 The proposed model

In this study, a new SCSOFS-HDL technique was introduced for the analysis of credit risks. The presented SCSOFS-HDL technique majorly focuses on the identification of good or bad credits. To do this, the SCSOFS-HDL method involves a four stage process namely data pre-processing, SCSOFS technique, DLSTM-SANN based classification, and PO based hyperparameter tuning. The working procedure is illustrated in Fig. 1.

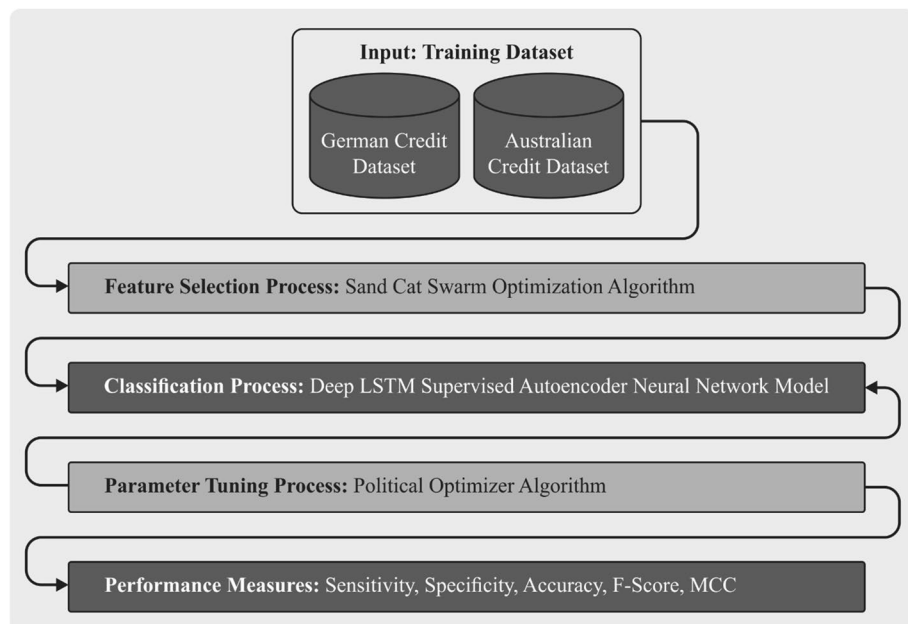


Fig. 1 Workflow of SCSOFS-HDL technique

3.1 Data pre-processing

At the primary level, the input credit risk data is pre-processed by the use of min-max normalization to scale the data into uniform format. Min-max normalization [23] was widely applied to scale the data s to unit variance. It is typically leveraged for computing the similarity degree between points. Assuming A as data, which was mapped from data range from A_{min} to A_{max} , using in Eq. (1):

$$A_{normalized} = \frac{A - A_{min}}{A_{max} - A_{min}} \quad (1)$$

The employment of min-max normalization ensured that the feature was exacted to similar scales.

3.2 Feature selection using SCSOFS technique

In this work, the normalized data is fed into the SCSOFS technique for effectual selection of features. The SCSO is a new metaheuristic optimization technique stimulated by survival behaviours of sand cats [24] and also they are capable of detecting lower frequency noise to find the prey below or above the ground. The presented technique takes the optimum values in exploration space as the prey, and through location update process, the search agent constantly explores the search space, ultimately moving towards the area where the optimum value is positioned. A prey attack strategy and a prey search process are the two strategies of SCSO method. The search prey mechanism simulates the sand cat's searching process for prey.

In the exploration stage, the sand cat senses lower frequency under $2kHz$. In the mathematical expression, r_G denotes a sensitivity range that reduces linearly within $[2,0]$ interval. S_M model the auditory property of dune cats with the assumption of 2, shown below [24].

$$\overline{r_G} = S_M - \left(\frac{S_M \times iter_c}{iter_{max}} \right) \quad (2)$$

The key parameter for the transformation between the development and exploration stage is R that which can be formulated by Eq. (3):

$$\vec{R} = 2 \times \overline{r_G} \times rand(0,1) - \overline{r_G} \quad (3)$$

The search space can be initialized at random between certain limitations. In the search process, every existing search agent's location can be upgraded according to the random location. Thereby, the search agent is capable of exploring new spaces in the search space. To prevent getting trapped in a local optimum, the sensitivity range for all the sand cats is not the same, as represented in Eq. (4):

$$\vec{r} = \overline{r_G} \rightarrow \times rand(0,1) \quad (4)$$

In this work, based on the optimum solution the sand cat update the existing location, and sensitivity range; search for other optimum prey location; and could discover a new optimum local in the search space, attaining a location between the prey and the existing

location, whereas the randomness guarantees that the presented method has lower complexity and running cost:

$$\overrightarrow{Pos}(t+1) = \vec{r} \cdot \left(\overrightarrow{Pos_b}(t) - rand(0,1) \cdot \overrightarrow{Pos_c}(t) \right) \quad (5)$$

In Eq. (7), Pos_b denotes the optimum solution, Pos_c shows the one's existing location, and r represents the sensitivity range. Following the prey search, The SCSO model attacks the prey, and this mechanism can be mathematically modelled as follows:

$$\overrightarrow{Pos_{md}} = |rand(0,1) \cdot \overrightarrow{Pos_b}(t) - \overrightarrow{Pos_c}(t)| \quad (6)$$

$$\overrightarrow{Pos}(t+1) = p_{os_b(t)}^{\rightarrow} - \vec{r} \cdot \overrightarrow{Pos_{md}} \cdot \cos(\theta) \quad (7)$$

From the expression, Pos_b denotes the optimal location, Pos_c represents the existing location, and Pos_{md} indicates the random location.

$$\overrightarrow{Pos}(t+1) = \begin{cases} p_{os_b(t)}^{\rightarrow} - \overrightarrow{Pos_{md}} \cdot \cos(\theta) \cdot \vec{r} & |R| \leq 1 \\ \vec{r}(p_{os_b(t)}^{\rightarrow} - rand(0,1) \cdot p_{os_c(t)}^{\rightarrow}) & |R| > 1 \end{cases} \quad (8)$$

In the exploitation and exploration phase, Eq. (10) characterizes the upgrade of every sand cat's location. If $R \leq 1$, the sand cat is guided for attacking the prey; or else, the sand cats are established by finding new potential solutions in the global region.

The fitness function of SCSO-FS method has considered the number of selective features and the classifier accuracy. It minimized the set size of the selected features and maximized the classifier accuracy. Thus, the following fitness function was utilized for assessing individual solutions, as seen in Eq. (9).

$$Fitness = \alpha * ErrorRate + (1 - \alpha) * \frac{\#SF}{\#All_F} \quad (9)$$

Here $\#All_F$ is the total attributes in the original data and $\#SF$ is the number of selected features. ErrorRate means the classifier error rate with the selective features. ErrorRate was computed as the percentage of incorrect classifiers to the total classifications made, expressed as a value between (0,1). α was utilized for controlling the significance of subset length and classification quality.

3.3 Credit risk analysis using DLSTM-SANN model

For automated credit risk prediction, the DLSTM-SANN model is used. The training of DLSTM-SANN model, shown diagrammatically in Fig. 2, is depending on the reduction of weighted summation of reconstructed loss function and supervised classification loss respective to the initial and second terms in (Eq. (13)) correspondingly [25]. The reduction of the classification loss makes sure the nonlinear hidden variable extracted is the predictor of the output labels whereas the reduction of the reconstructed loss function in Eq. (13) makes sure that the evaluated hidden parameter can capture the variance in the input dataset. The mean squared error function can be utilized as a reconstructed loss and softmax cross-entropy as classification loss. The main objective is to learn the function which forecast class labels in one hot encoded form $y_j \in \mathbb{R}^m$ from the input $x_j \in \mathbb{R}^{d_i \times 1}$. In the presented method, LSTM unit is used rather than dense layer for the encoding and decoding function. The objective is

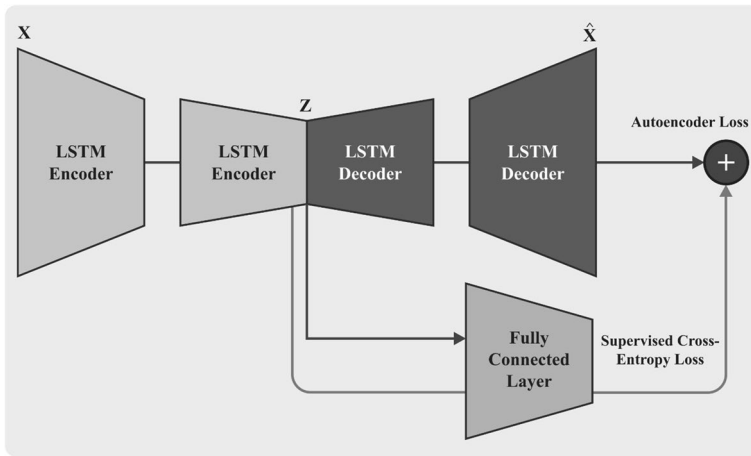


Fig. 2 Structure of DLSTM-SANN model

to recreate and simultaneously categorize the input sequence at t time. The encoding function transforms the input time sequence data to learn significant features and encode the feature $z \in \mathbb{R}^{d_0 \times 1}$. The decoding function recreates the input utilizing the extracted feature vector.

The process implemented by the encoding function for single LSTM layer between the inputs to the hidden parameters $z_t^i \in \mathbb{R}^{d_0 \times 1}$ can be formulated below:

$$z_t^i = \zeta_e(x_t^i) \quad (10)$$

Equation (10) is the LSTM encoding function. The hidden parameter is used for predicting the class label. Where ζ_e denotes the LSTM encoded function. The hidden parameters are used to reconstruct the input x and to predict the class labels as follows [25]:

$$x_t^i \wedge = \zeta_d(z_t^i) \quad (11)$$

$$\hat{y}_t^i = f_c(W_c z_t^i + b_c) \quad (12)$$

Where $W_c \in \mathbb{R}^{m \times d_c}$ and $b_c \in \mathbb{R}^m$ indicates output weight matrix and bias vector correspondingly. f_c denotes a nonlinear activation function employed for the output layer. The subsequent loss function can be minimized for training the SAE:

$$\begin{aligned} l_{DSAE} &= \frac{\lambda_1}{N} \|x_s - \hat{x}_s\|_2^2 + \frac{1}{N} \sum_{s=1}^N \sum_{c=1}^m -y_{s,c} \log(p_{s,c}) \\ &= \frac{1}{N} \left[\lambda_1 \|x_s - \hat{x}_s\|_2^2 + \sum_{s=1}^N \sum_{c=1}^m -y_{s,c} \log(p_{s,c}) \right] \end{aligned} \quad (13)$$

$$p_{s,c} = \frac{e^{(y_{s,c}^\wedge)}}{\sum_{c=1}^m e^{(y_{s,c}^\wedge)}} \quad (14)$$

Now, $y_{s,c}$ shows the binary indicator (zero or one) equivalent to one when the class label c is the accurate one for observation s and 0 if not, λ_1 denotes weight multiplying the reconstructed loss L_r in the cost that should be minimalized, m shows the quantity of classes, $p_{s,c}$ indicates the prediction probability for the s samples of class c and $y_{s,c}^\wedge$ denotes the non-normalized log probability.

Furthermore, to prevent overfitting, add a regularization term to the objective function in Eq. (13). Therefore, the mathematical expression for DLSTM-SANN is represented as follows:

$$\min l_{DSAE} = \min \frac{1}{N} \left[\lambda_1 \|x_s - \hat{x}_s\|_2^2 + \lambda_2 \sum_{s=1}^N \sum_{c=1}^m -y_{s,c} \log(p_{s,c}) + \lambda_3 \sum_L \sum_k \sum_j W_{kj^2}^{[L]} \right] \quad (15)$$

In Eq. (15), $W_{kj}^{[L]}$ indicates the weight matrix for all the layers L in network and the weight on individual objective function $\lambda_1, \lambda_2, \lambda_3$ are selected by the validation dataset.

3.4 Hyperparameter tuning using PO algorithm

Lastly, the PO algorithm is used for the optimum parameter tuning of the DLSTM-SANN model. PO is a new intellectual optimization technique stimulated by political election behavior of human society [26]. In this work, every party member can be seen as a solution candidate, and the election process of party members is regarded as an assessment function. Furthermore, the vote attained by party members was mapped onto the fitness value of the solution candidate. Different from the conventional techniques, PO considers the overall process of political election, along with election campaign, parliamentary affairs, inter-party election, party switching, party formation and constituency allocation based on political election. PO search for the optimum solution through a multi-phase iteration technique. The five main stages of *PO* are discussed in the following.

Party formation and constituency allocation In the initial stage, the overall population has n^2 members were split into n parties, and there exist n individuals (solution candidates) in every party. Moreover, all the party members perform the role of election candidate, viz., to form a constituency, one member from every party should be carefully chosen, the blue dotted line shows the division of constituency, and the red dotted line specifies division of political parity. The mapping of population division to numerical expression is that the overall population was split into n political parties, and every party includes n party members as follows [27]:

$$P = \{P_1, P_2, P_3, \dots, P_n\} \quad (16)$$

$$P_j = \{p_i^1, p_i^2, p_i^3, \dots, p_i^n\} \quad (17)$$

Also, every party member has played the part of election candidate, as represented in Eq. (18) the overall population is considered as n constituencies. What should be highlighted is the member of the constituency are party members, however, the logical division is not the same. As demonstrated in Eq. (19), the membership of every constituency is divided.

$$C = \{C_1, C_2, C_3, \dots, C_n\} \quad (18)$$

$$C_j = \{p_1^j, p_2^j, p_3^j, \dots, p_n^j\} \quad (19)$$

Moreover, the leader of i -th parties afterwards calculating the fitness of every member is perceived as p_i^* and the group of each party leader can be denoted as P^* as demonstrated in Eq. (20). Likewise, after the election, C^* regroup the winner from every constituency termed the parliamentarian where c_j^* shows the winner of j -th constituencies [27], as given below.

$$P^* = \{p_1^*, p_2^*, p_3^*, \dots, p_n^*\} \quad (20)$$

$$C^* = \{c_1^*, c_2^*, c_3^*, \dots, c_n^*\} \quad (21)$$

Election campaign This is the key phase of the PO and is accountable for the position updating of the searching agents. In this work, the party member changes their position based on the leader p^* of the party they belong to and the winner C^* of the constituency. Furthermore, they benefit from the experience acquired from last election through a novel position update process named current past-based position updating strategy (RPPUS), as expressed in Eqs. (22) and (23). The major concept is to forecast potential areas using the mathematical relation between subgroup optimum solution (constituency winner or party leader) and present and prior fitness of searching agent.

$$p_{i,k}^j(t+1) = \begin{cases} m^* + r(m^* - p_{i,k}^j(t)), & \text{if } p_{i,k}^j(t-1) \leq \mu_k(t) \leq m^* \text{ or } p_{i,k}^j(t-1) \geq \mu_k(t) \geq m^* \\ m^* + (2r-1)|m^* - p_{i,k}^j(t)|, & \text{if } p_{i,k}^j(t-1) \leq m^* \leq p_{i,k}^j(t) \text{ or } p_{i,k}^j(t-1) \geq m^* \geq p_{i,k}^j(t) \\ m^* + (2r-1)|m^* - p_{i,k}^j(t-1)|, & \text{if } m^* \leq p_{i,k}^j(t-1) \leq p_{i,k}^j(t) \text{ or } m^* \geq p_{i,k}^j(t-1) \geq p_{i,k}^j(t) \end{cases} \quad (22)$$

$$p_{i,k}^i(t+1) = \begin{cases} m^* + (2r-1)|m^* - p_{i,k}^i(t)|, & \text{if } p_{i,k}^i(t-1) \leq p_{i,k}^i(t) \leq m^* \text{ or } p_{i,k}^i(t-1) \geq p_{i,k}^i(t) \geq m^* \\ p_{i,k}^i(t-1) + r(p_{i,k}^i(t) - p_{i,k}^i(t-1)), & \text{if } p_{i,k}^i(t-1) \leq m^* \leq p_{i,k}^i(t) \text{ or } p_{i,k}^i(t-1) \geq m^* \geq p_{i,k}^i(t) \\ m^* + (2r-1)|m^* - p_{i,k}^i(t-1)|, & \text{if } m^* \leq p_{i,k}^i(t-1) \leq p_{i,k}^i(t) \text{ or } m^* \geq p_{i,k}^i(t-1) \geq p_{i,k}^i(t) \end{cases} \quad (23)$$

Where m^* shows the winner of the constituency or the leader of party, r characterizes a random number from zero to one, and t signifies existing iteration number.

Party switching This stage balances exploitation and exploration that present an adaptive variable named λ party switching rate. Every party member is chosen and swapped to randomly selected party. The switching probability can be defined by λ was initially 1 and linearly reduced to zero as demonstrated in Eq. (24).

$$\lambda = (1 - \frac{t}{T}) * \lambda_{\max} \quad (24)$$

Election. In this phase, the fitness of every candidate solution was defined and the constituency winner and party leader are upgraded using the following expression:

$$q = \operatorname{argmin}_{i \leq n} (p_i^j) p_i^* = p_i^q \quad (25)$$

$$q = \operatorname{argmin}_{i \leq n} (p_i^j) c_j^* = p_j^q \quad (26)$$

The PO approach has derived a fitness function to get better classifier performance. It determined a positive integer that designated superior outcome of the solution candidate. Here, the reduction of classifier error rate is the fitness function, as shown in Eq. (27).

$$\text{fitness}(x_i) = \text{Classifier Error Rate}(x_i) = \frac{\text{number of misclassified samples}}{\text{Total number of samples}} * 100 \quad (27)$$

4 Results and discussion

The proposed model is simulated using Python 3.8.5 tool on PC i5-8600k, GeForce 1050Ti 4GB, 16GB RAM, 250GB SSD, and 1 TB HDD. In this study, the credit risk investigation of the SCSOFS-HDL technique is studied on two benchmark datasets from UCI repository. The first German Credit dataset ([https://archive.ics.uci.edu/ml/datasets/statlog+\(german+credit+data\)](https://archive.ics.uci.edu/ml/datasets/statlog+(german+credit+data))) includes 1000 samples with 24 attributes (Selected attributes are 16) and the second Australian Credit dataset ([https://archive.ics.uci.edu/ml/datasets/statlog+\(australian+credit+approval\)](https://archive.ics.uci.edu/ml/datasets/statlog+(australian+credit+approval))) comprises 690 samples with 14 attributes (Selected attributes are 9). Table 1 provides the details of dataset used.

Table 2 provides an overall best cost (BC) examination of the SCSO algorithm for FS purposes. The results are studied under the execution of 20 iterations. The experimental values pointed out that the SCSO system reaches optimal values on both datasets.

The actual classification results of the SCSOFS-HDL technique can be described in the confusion matrix on German Credit dataset, as illustrated in Fig. 3. The results indicate that the SCSOFS-HDL technique predicts credit and non credit cases in a most probable way. In addition, the results notify that the SCSOFS-HDL technique led to consistent results under different training phase (TRP) and testing phase (TSP) aspects.

To examine the credit assessment performance of the SCSOFS-HDL technique on German credit dataset, overall outcomes are briefed in Table 3; Fig. 4. The outcomes implied that the SCSOFS-HDL technique can efficaciously identify the credit and non-credit samples on all TRP and TSP cases. For instance, under 70% of TRP, the SCSOFS-HDL technique gains average $accu_{bal}$ of 95.43%, $sens_y$ of 95.43%, $spec_y$ of 95.43%, F_{score}

Table 1 Dataset used

Class	Dataset	
	German Credit	Australian Credit
Credit Approval	700	307
Non Credit Approval	300	383
Total Number of Instances	1000	690

Table 2 Comparative BC analysis of SCSOFS method

Best Cost Analysis on Applied Dataset

No. of Iterations	German Credit Dataset				Australian Credit Dataset			
	PSO	GWO	ACO	SCSO	PSO	GWO	ACO	SCSO
1	0.1787	0.1894	0.1650	0.1420	0.1029	0.1078	0.0918	0.0681
2	0.1741	0.1876	0.1662	0.1459	0.0983	0.1057	0.0928	0.0775
3	0.1748	0.1673	0.1681	0.1399	0.1027	0.1085	0.0888	0.0650
4	0.1761	0.1693	0.1659	0.1507	0.1013	0.1071	0.0906	0.0721
5	0.1729	0.1654	0.1586	0.1343	0.0988	0.1036	0.0922	0.0701
6	0.1756	0.1658	0.1560	0.1356	0.0988	0.1012	0.0905	0.0606
7	0.1749	0.1657	0.1576	0.1311	0.0961	0.1035	0.0923	0.0630
8	0.1736	0.1675	0.1476	0.1347	0.0977	0.1014	0.0897	0.0691
9	0.1725	0.1672	0.1477	0.1248	0.0983	0.1022	0.0905	0.0762
10	0.1741	0.1660	0.1497	0.1287	0.0962	0.1028	0.0912	0.0747
11	0.1748	0.1694	0.1465	0.1324	0.0964	0.1032	0.0882	0.0659
12	0.1714	0.1688	0.1465	0.1365	0.0970	0.0996	0.0877	0.0624
13	0.1749	0.1663	0.1481	0.1271	0.0964	0.0927	0.0867	0.0644
14	0.1737	0.1677	0.1479	0.1316	0.1000	0.0941	0.0888	0.0727
15	0.1705	0.1668	0.1468	0.1277	0.0981	0.0923	0.0885	0.0653
16	0.1720	0.1682	0.1484	0.1328	0.0976	0.0946	0.0869	0.0618
17	0.1703	0.1653	0.1492	0.1343	0.0996	0.0956	0.0870	0.0646
18	0.1718	0.1674	0.1496	0.1378	0.0974	0.0960	0.0855	0.0688
19	0.1661	0.1687	0.1480	0.1357	0.0981	0.0928	0.0858	0.0695
20	0.1685	0.1692	0.1457	0.1243	0.0971	0.0941	0.0866	0.0640

of 96.42%, and MCC of 93.01%. Next, with 30% of TSP, the SCSOFS-HDL technique receives average $accu_{bal}$ of 96.49%, $sens_y$ of 96.49%, $spec_y$ of 96.49%, F_{score} of 97.03%, and MCC of 94.09%.

Figure 5 examines the accuracy of the SCSOFS-HDL technique during the training and validation process on German Credit dataset. The figure notifies that the SCSOFS-HDL technique reaches increasing accuracy values over increasing epochs. In addition, the increasing validation accuracy over training accuracy exhibits that the SCSOFS-HDL technique learns efficiently on the German Credit dataset.

The loss analysis of the SCSOFS-HDL technique at the time of training and validation is demonstrated on the German Credit dataset in Fig. 6. The outcomes specify that the SCSOFS-HDL technique reaches closer values of training and validation loss. It is observed that the SCSOFS-HDL technique learns efficiently on the German Credit dataset.

The loss analysis of the SCSOFS-HDL technique at the time of training and validation is demonstrated on the German Credit dataset in Fig. 6. The outcomes designate that the SCSOFS-HDL technique reaches closer values of training and validation loss. It is observed that the SCSOFS-HDL technique learns efficiently on the German Credit dataset.

The actual classification results of the SCSOFS-HDL method were explained in the confusion matrix on Australian Credit dataset, as seen in Fig. 7. The results indicate that

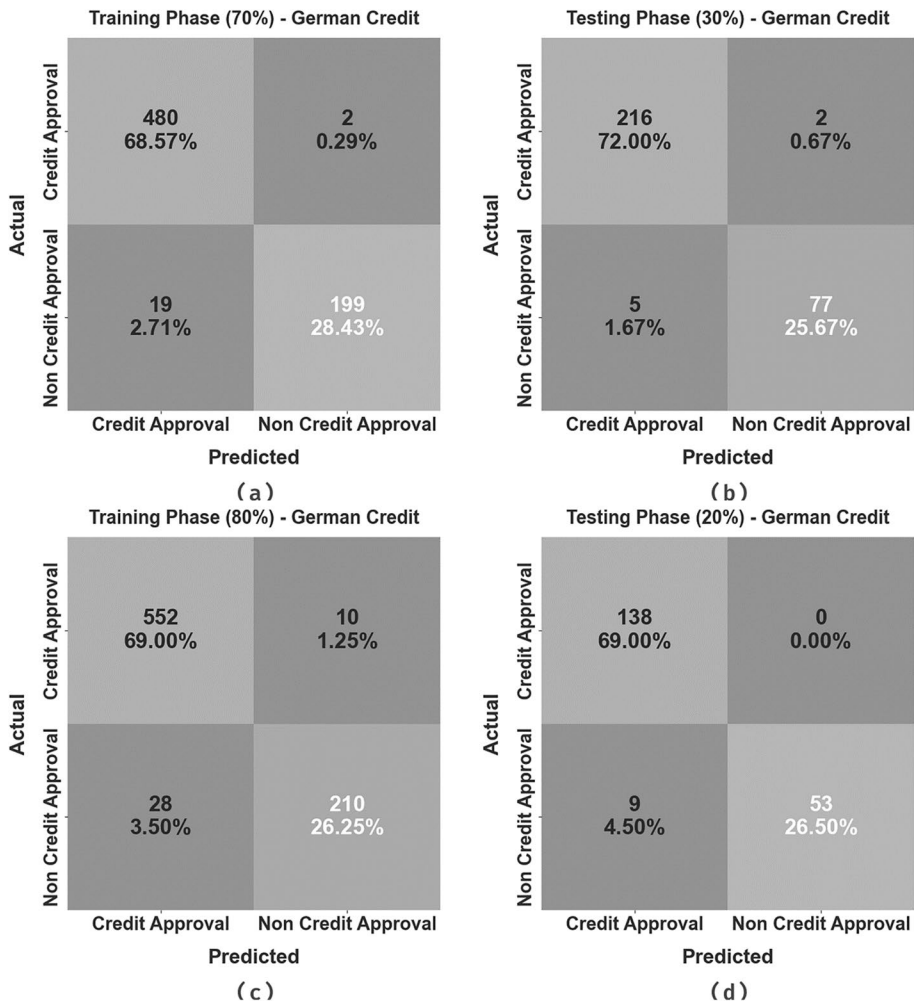


Fig. 3 Confusion matrices of SCSOFS-HDL technique on German Credit dataset

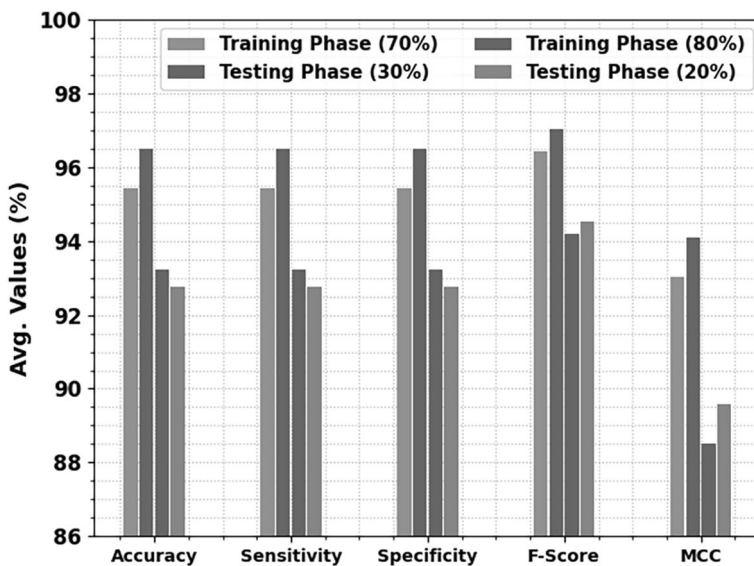
the SCSOFS-HDL method predicts credit and non credit cases in a most probable way. As well, the results notify that the SCSOFS-HDL technique led to consistent results under different TRP and TSP aspects.

To examine the credit assessment performance of the SCSOFS-HDL technique on Australian credit dataset, the overall results are briefed in Table 4; Fig. 8. The outcomes implied that the SCSOFS-HDL technique can efficaciously identify the credit and non-credit samples on all TRP and TSP cases. For instance, under 70% of TRP, the SCSOFS-HDL technique gains average $accu_{bal}$ of 95.98%, $sens_y$ of 95.98%, $spec_y$ of 95.98%, F_{score} of 96.37%, and MCC of 92.88%. Next, with 30% of TSP, the SCSOFS-HDL technique receives average $accu_{bal}$ of 96.12%, $sens_y$ of 96.12%, $spec_y$ of 96.12%, F_{score} of 96.13%, and MCC of 92.28%.

Figure 9 examines the accuracy of the SCSOFS-HDL technique during the training and validation process on Australian Credit dataset. The figure notifies that the SCSOFS-HDL

Table 3 Credit result analysis of SCSOFS-HDL technique on German Credit dataset

Class	$Accu_{bal}$	$Sens_y$	$Spec_y$	F_score	MCC
Training Phase (70%)					
Credit Approval	99.59	99.59	91.28	97.86	93.01
Non Credit Approval	91.28	91.28	99.59	94.99	93.01
Average	95.43	95.43	95.43	96.42	93.01
Testing Phase (30%)					
Credit Approval	99.08	99.08	93.90	98.41	94.09
Non Credit Approval	93.90	93.90	99.08	95.65	94.09
Average	96.49	96.49	96.49	97.03	94.09
Training Phase (80%)					
Credit Approval	98.22	98.22	88.24	96.67	88.52
Non Credit Approval	88.24	88.24	98.22	91.70	88.52
Average	93.23	93.23	93.23	94.19	88.52
Testing Phase (20%)					
Credit Approval	100.00	100.00	85.48	96.84	89.58
Non Credit Approval	85.48	85.48	100.00	92.17	89.58
Average	92.74	92.74	92.74	94.51	89.58

**Fig. 4** Overall Credit result analysis of SCSOFS-HDL technique on German Credit dataset

technique reaches increasing accuracy values over increasing epochs. In addition, the increasing validation accuracy over training accuracy exhibits that the SCSOFS-HDL technique learns efficiently on the Australian Credit dataset.

The loss analysis of the SCSOFS-HDL technique at the time of training and validation is demonstrated on the Australian Credit dataset in Fig. 10. The outcomes indicate that the SCSOFS-HDL technique reaches closer values of training and validation loss. It

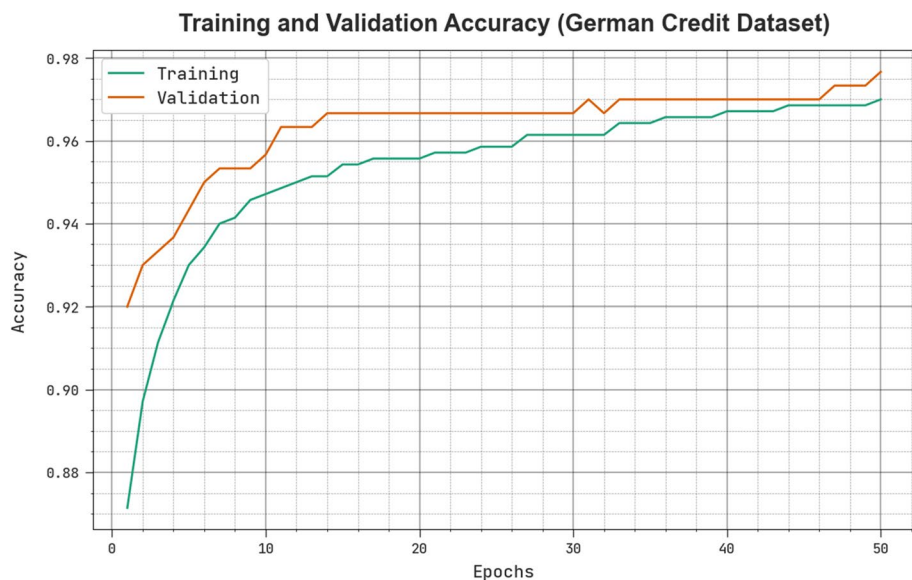


Fig. 5 Accuracy curves of the SCSOFS-HDL technique on German Credit dataset

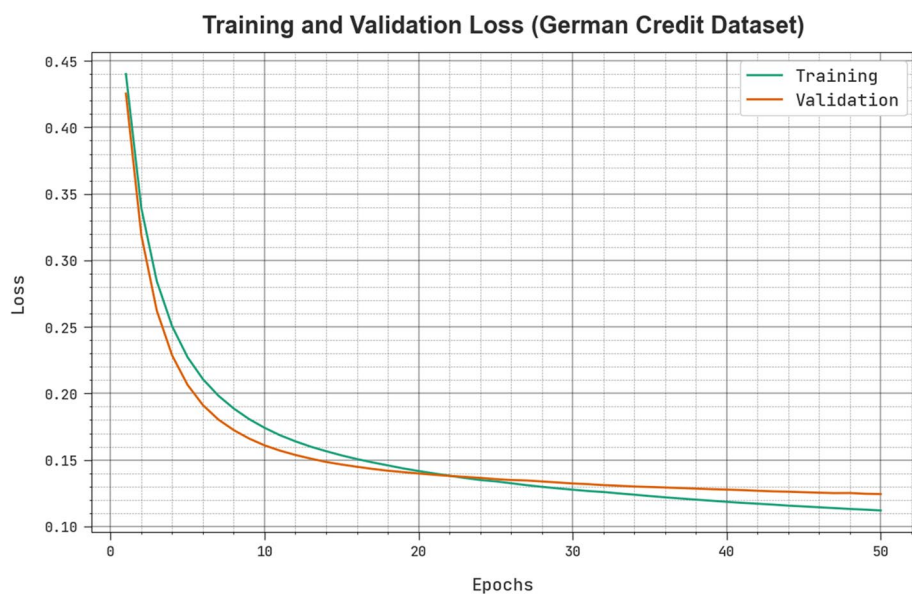


Fig. 6 Loss curves of the SCSOFS-HDL approach on German Credit dataset

is observed that the SCSOFS-HDL technique learns efficiently on the Australian Credit dataset.

Table 5 demonstrates detailed comparative credit risk predictive results of the SCSOFS-HDL technique with other existing classifiers [28, 29]. Figure 11 represents a comparison

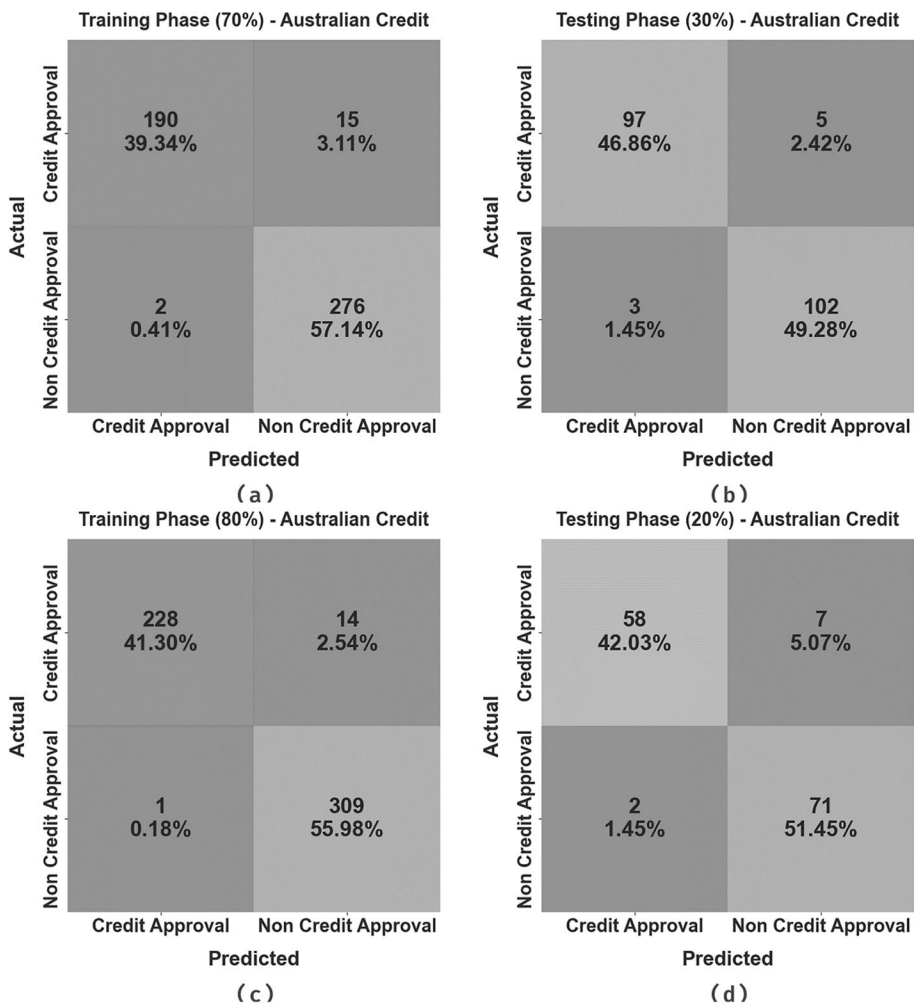


Fig. 7 Confusion matrices of SCSOFS-HDL technique on Australian Credit dataset

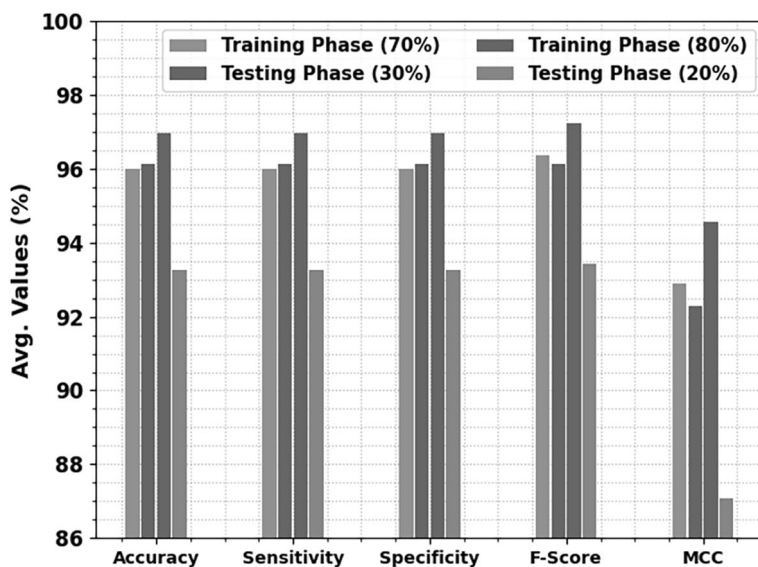
study of the SCSOFS-HDL technique in terms of $sens_y$ and $spec_y$. The results indicate that the MLP, AdaBoost, and Olex-GA model shows worse predictive outcomes than other recent models. Contrastingly, the NB-RT model exhibits reasonable performance with $sens_y$ and $spec_y$ of 93.64% and 74.98%. Although the DL-fine tuned and DNN models resulted in near optimal values of $sens_y$ and $spec_y$, the SCSOFS-HDL technique outperforms the existing ones with maximum $sens_y$ and $spec_y$ of 96.95% and 96.95% respectively.

Figure 12 represents a comparison study of the SCSOFS-HDL technique in terms of $accu_y$ and F_{score} . The results indicate that the MLP, AdaBoost, and Olex-GA model shows worse predictive outcomes than other recent models. Contrastingly, the NB-RT model exhibits reasonable performance with $accu_y$ and F_{score} of 91.31% and 94.98%.

Although the DL-fine tuned and DNN models resulted in near optimal values of $accu_y$ and F_{score} , the SCSOFS-HDL technique outperforms the existing ones with maximum $accu_y$ and F_{score} of 96.95% and 97.22% respectively. From these results, it can be confirmed

Table 4 Credit result analysis of SCSOFS-HDL technique on Australian Credit dataset

Class	Accuracy _{bal}	Sensitivity	Specificity	F-Score	MCC
Training Phase (70%)					
Credit Approval	92.68	92.68	99.28	95.72	92.88
Non Credit Approval	99.28	99.28	92.68	97.01	92.88
Average	95.98	95.98	95.98	96.37	92.88
Testing Phase (30%)					
Credit Approval	95.10	95.10	97.14	96.04	92.28
Non Credit Approval	97.14	97.14	95.10	96.23	92.28
Average	96.12	96.12	96.12	96.13	92.28
Training Phase (80%)					
Credit Approval	94.21	94.21	99.68	96.82	94.56
Non Credit Approval	99.68	99.68	94.21	97.63	94.56
Average	96.95	96.95	96.95	97.22	94.56
Testing Phase (20%)					
Credit Approval	89.23	89.23	97.26	92.80	87.09
Non Credit Approval	97.26	97.26	89.23	94.04	87.09
Average	93.25	93.25	93.25	93.42	87.09

**Fig. 8** Overall Credit result analysis of SCSOFS-HDL technique on Australian Credit dataset

that the SCSOFS-HDL technique accomplishes promising results over other existing models.

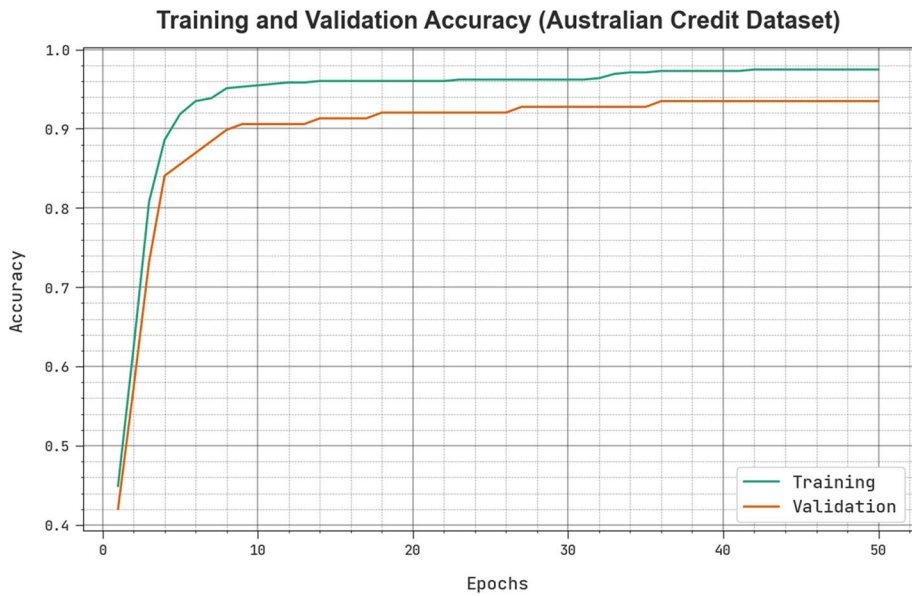


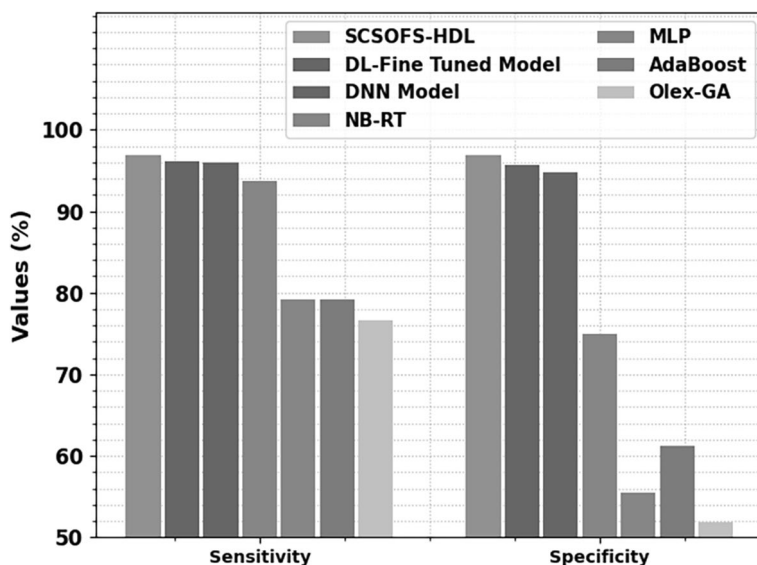
Fig. 9 Accuracy curves of the SCSOFS-HDL technique on Australian Credit dataset



Fig. 10 Loss curves of the SCSOFS-HDL technique on Australian Credit dataset

Table 5 Comparative results of SCSOFS-HDL technique with other classifiers [28, 29]

Methods	Sensitivity	Specificity	Accuracy	F-score
SCSOFS-HDL	96.95	96.95	96.95	97.22
DL-Fine Tuned Model	96.11	95.65	96.24	96.52
DNN Model	96.04	94.83	94.96	96.32
NB-RT	93.64	74.98	91.31	94.98
MLP	79.15	55.44	74.37	81.05
AdaBoost	79.21	61.19	75.99	83.31
Olex-GA	76.62	51.92	71.20	80.42

**Fig. 11** Comparative $sens_y$ and $spec_y$ results of SCSOFS-HDL technique

5 Conclusion

In this study, a new SCSOFS-HDL algorithm was introduced for the analysis of credit risks. The presented SCSOFS-HDL technique majorly focuses on the identification of good or bad credits. To accomplish this, the SCSOFS-HDL technique involves data pre-processing, SCSOFS technique, DLSTM-SANN based classification, and PO based hyperparameter tuning. The design of the SCSOFS technique lies in the effectual identification of features and thereby results in enhanced classification outcomes. At the same time, the inclusion of PO based parameter tuning process leads to the betterment of the SCSOFS technique. The experimental validation of the SCSOFS-HDL technique is tested on German Credit and Australian Credit datasets. The experimental results highlighted the better performance of the SCSOFS-HDL technique compared to

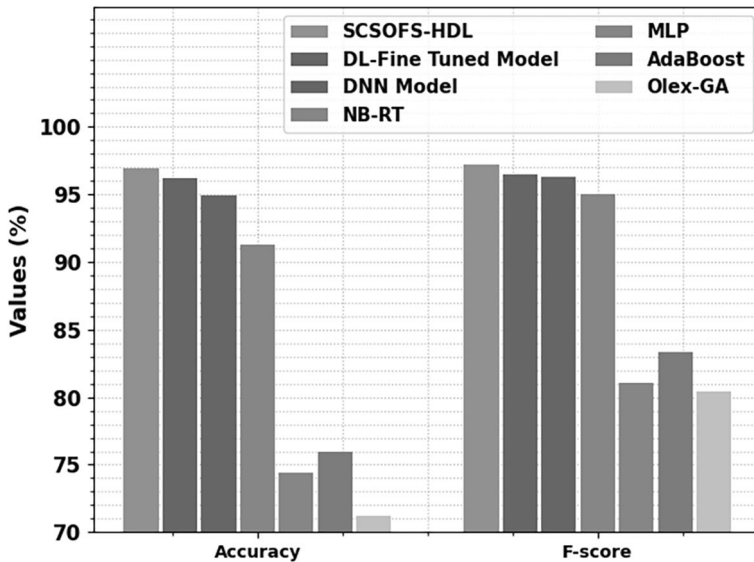


Fig. 12 Comparative $accu_y$ and F_{score} , results of SCSOFS-HDL technique

existing credit risk models with maximum accuracy of 96.49% and 96.12% on German Credit and Australian Credit datasets, respectively. Therefore, the proposed model can be applied for accurate and automated credit risk assessment process. In future, outlier detection models can enhance the potential of the SCSOFS-HDL technique. In addition, robust credit risk assessment models should be developed to handle class imbalance datasets. In addition, real-time fraud detection capabilities can be improved by leveraging edge computing, stream processing, and event-driven architectures to make instant decisions and prevent fraudulent transactions in real-time environment.

Funding The authors received no funding for this study.

Data availability Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethics approval Not Applicable.

Human participants and/or animals Not applicable.

Consent to participate Not applicable.

Conflict of interest The authors declare that they have no conflict of interest. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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